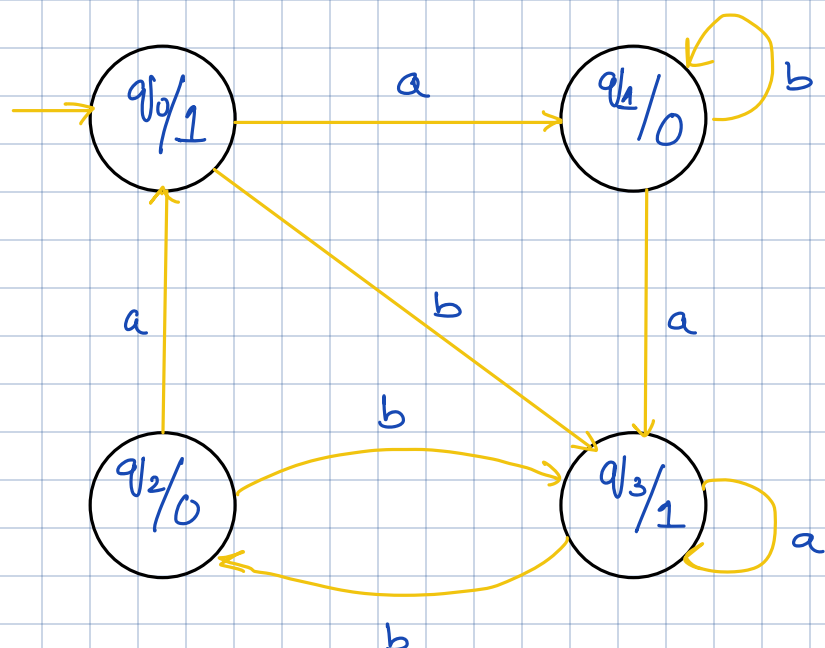
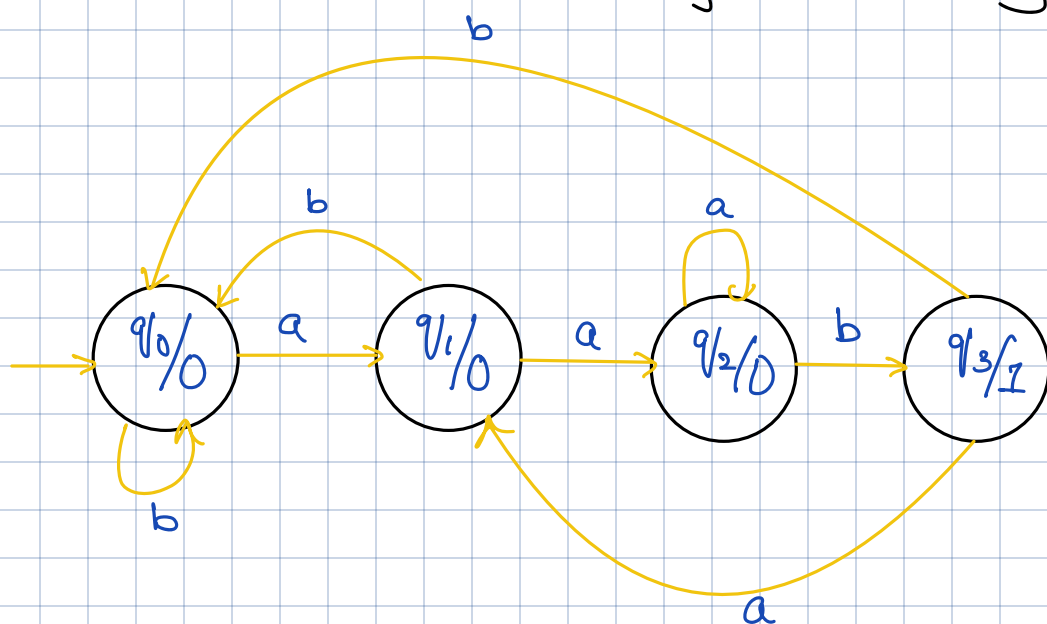


Moose's Machine:

- ↳ FA that has outputs on the state
- ↳ Such machines have no concept of acceptance or rejection



Example, Count occurrences of sub-string 'aab'

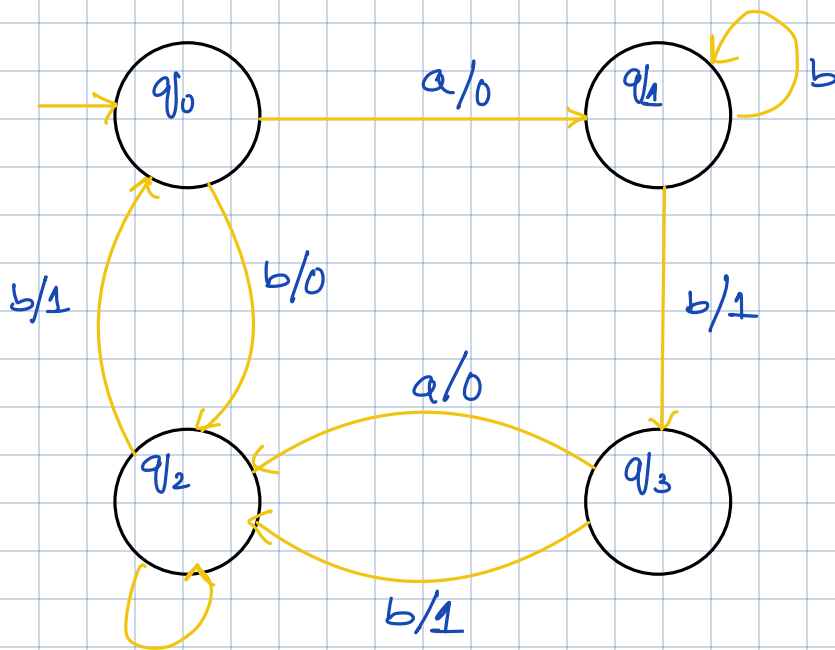


↳ On q_0 state at start output what is at q_0 .

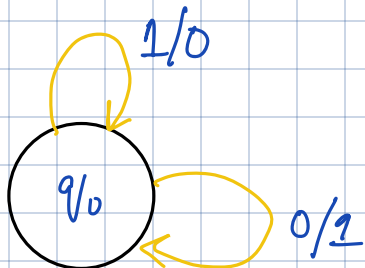
Mealy Machine:

↳ Have outputs on transitions

↳ Output of Moore is one more than mealy machine.

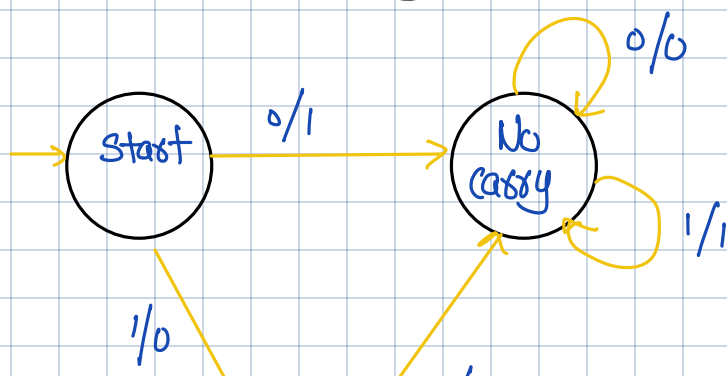


For example, Invest input

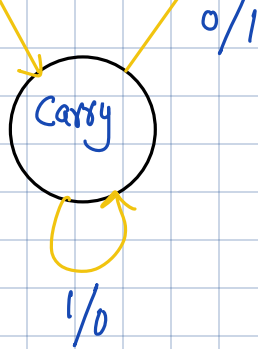


Build a machine to:

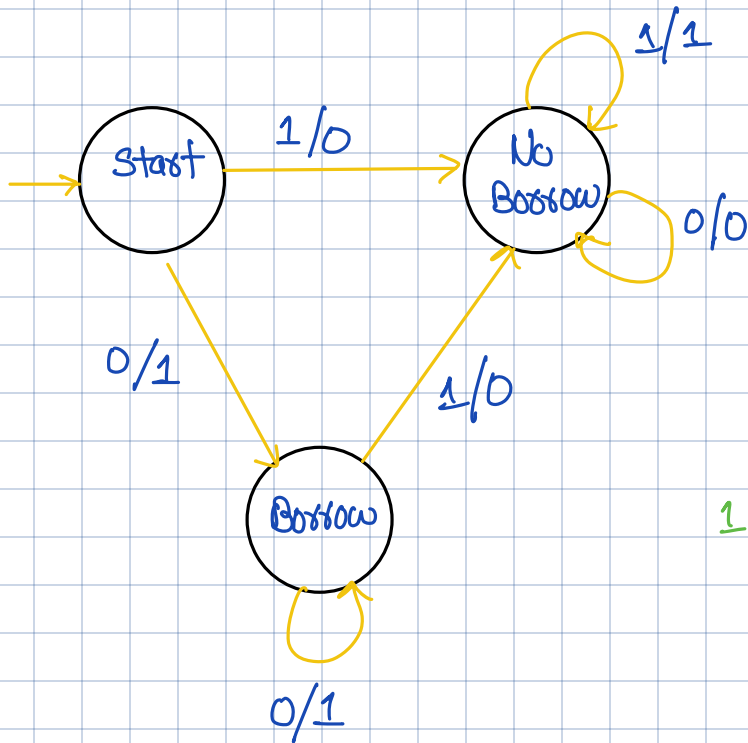
1. Increment by 1



1	1		
1	0	1	1
			1
1	1	0	0



2. Decrement by 1

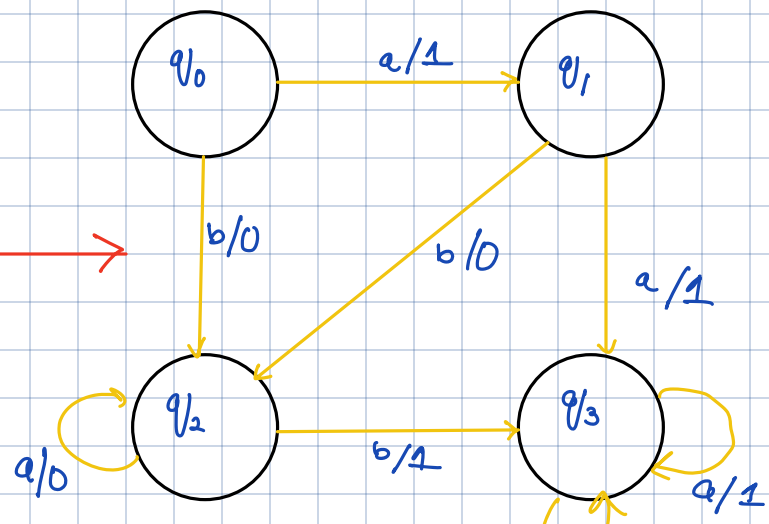
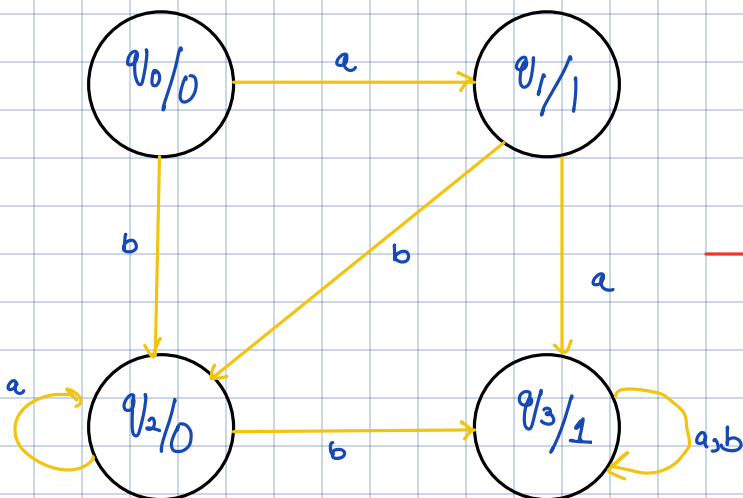


$$\begin{array}{r} 1101 \\ - \quad 1 \\ \hline 1100 \end{array} \quad \begin{array}{r} 0110 \\ 1010 \\ \hline 1011 \end{array}$$

1011

Moore to Mealy

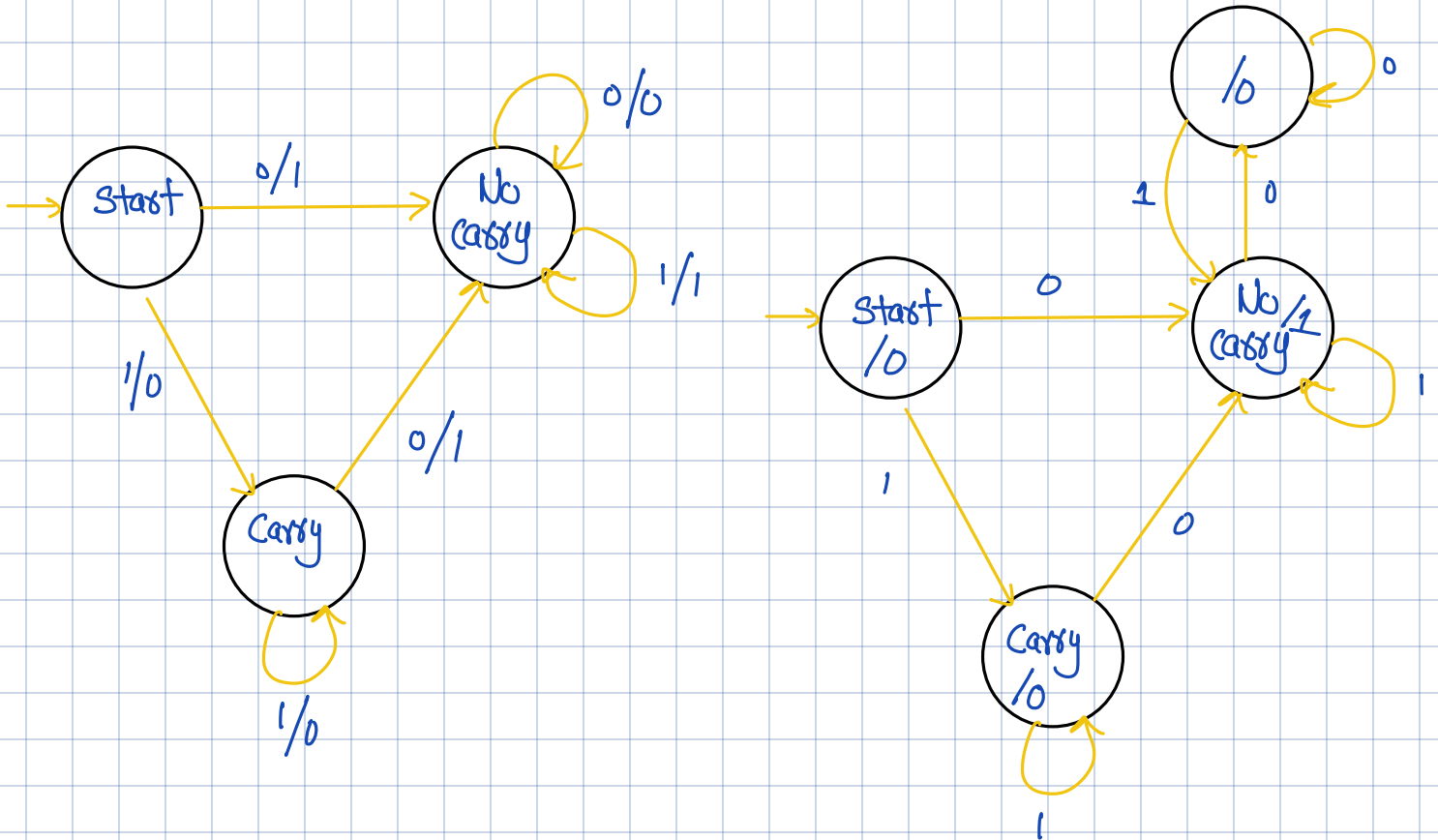
- ↳ Move outputs onto incoming edges.
- ↳ Leave out-going edges alone



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Mealy to Moore

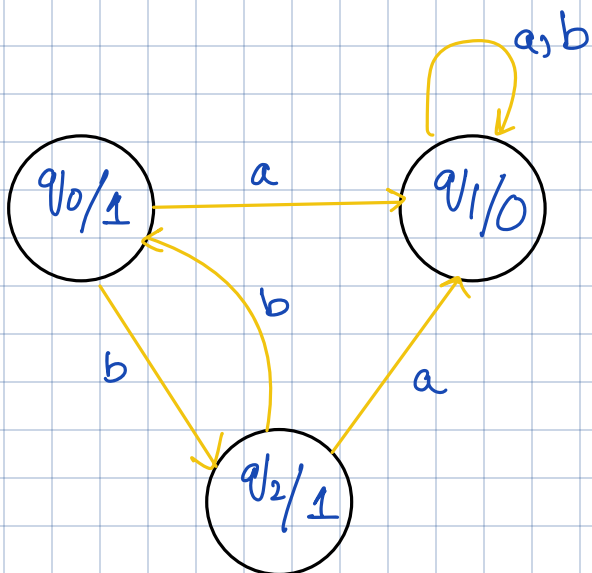
- ↳ Group common output transitions on a new state
- ↳ Complete transitions



Exercise:

Q1.

(i)



Q3.

(i)

	a	b	output
q ₀	q ₀	q ₁	0
q ₁	q ₁	q ₁	1

Q6.

(i)

	q ₀	q ₁
q ₀	a/0	b/1
q ₁	a/1	b/0

Q4.

Q1(i) aabab

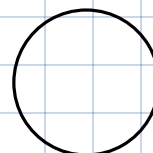
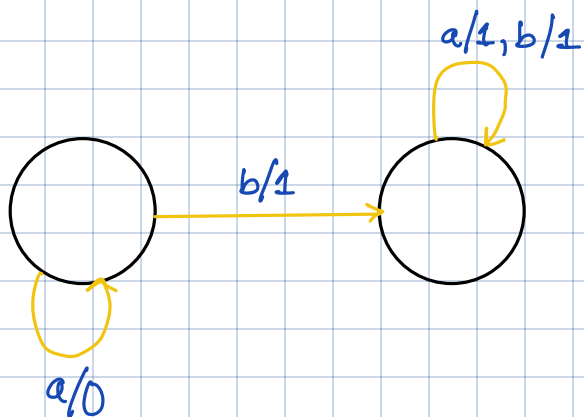
100000

Q3.(i)

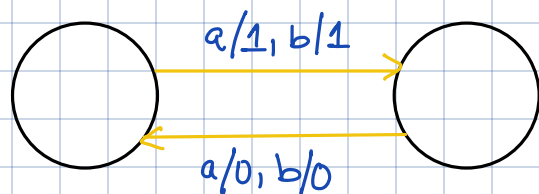
000111

Q7.

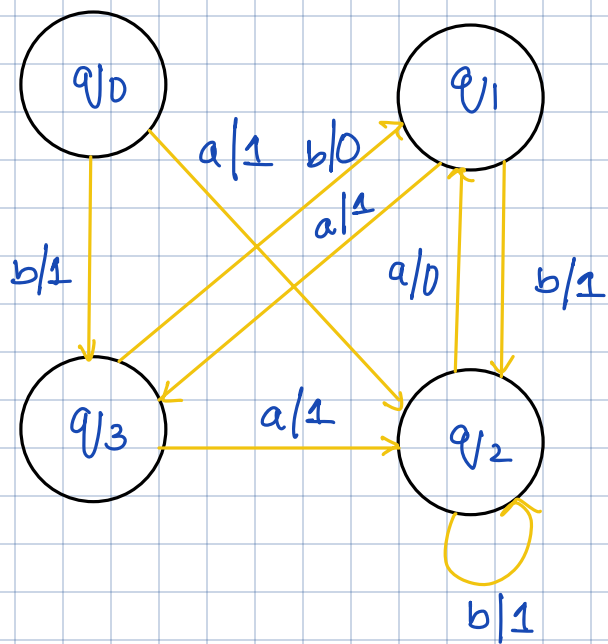
(i)



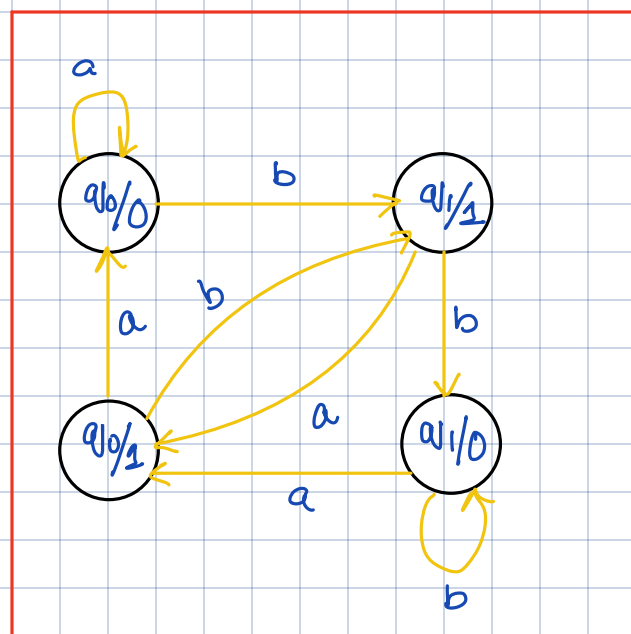
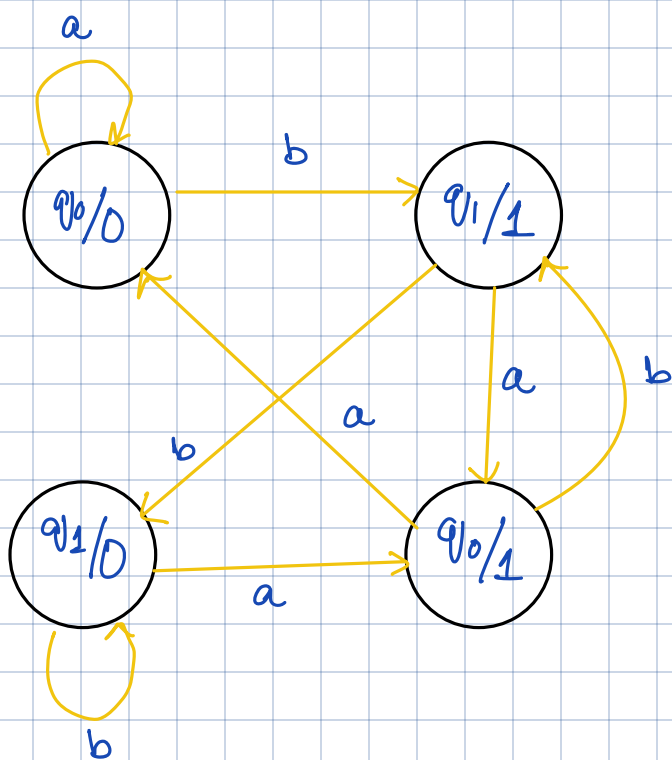
(ii)



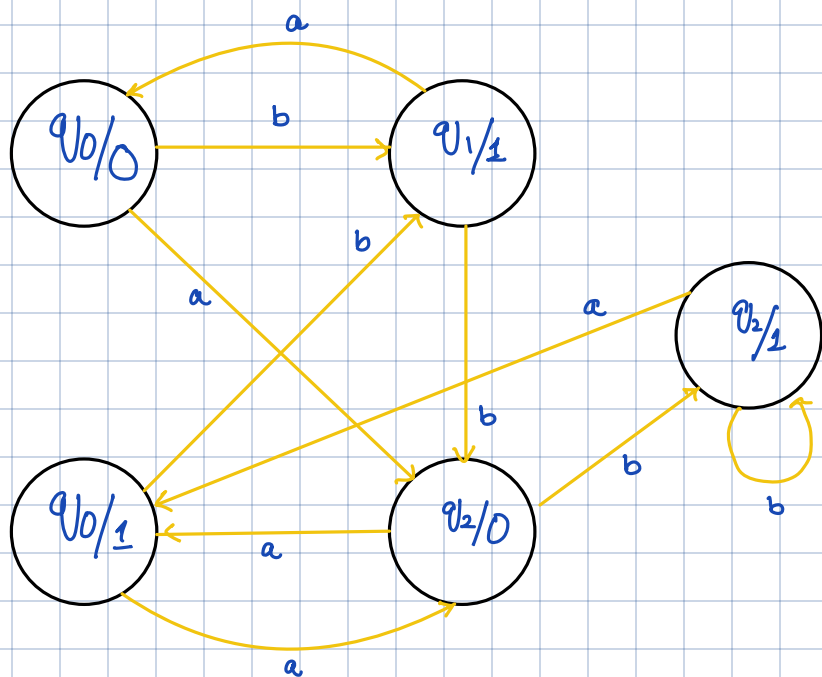
(v)



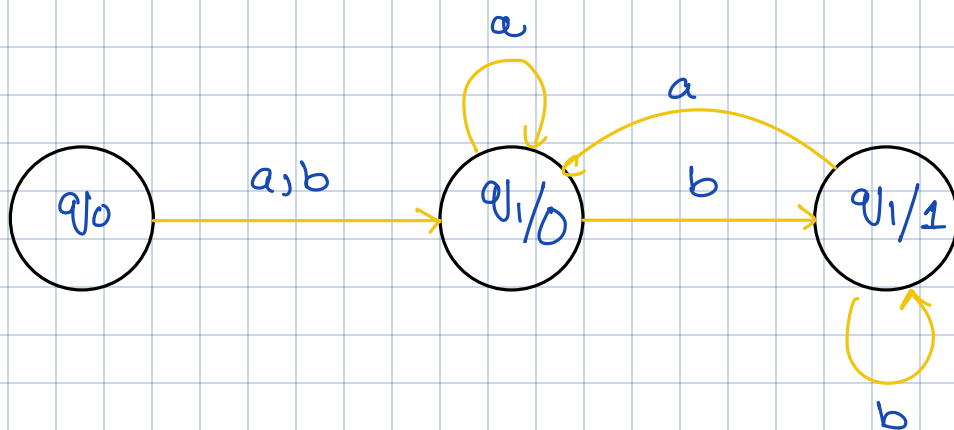
Q_{q_0}
(i)



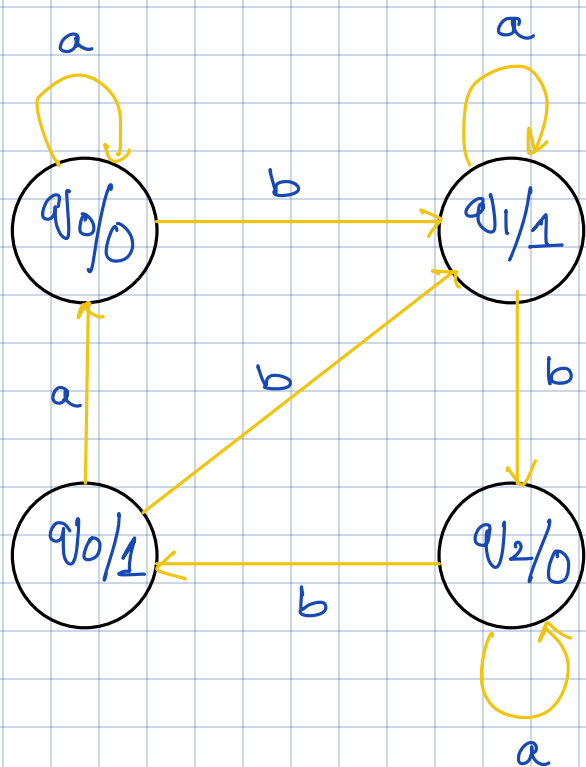
(iii)



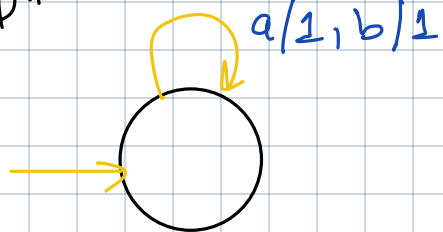
(iii)



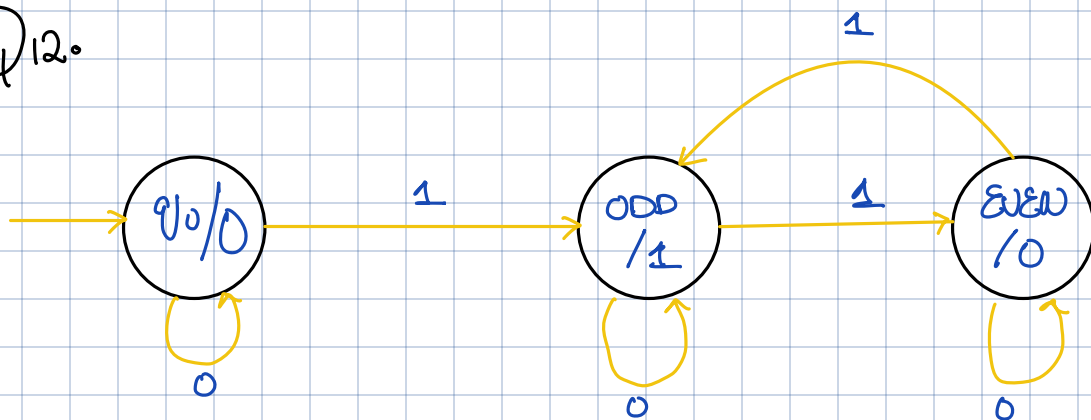
(iv)



Q11.



Q12.



OR

